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Wind-resistant performances comparison between a long-span CFRP and a steel cable-stayed

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Abstract

Two ultra-long-span cable-stayed bridge schemes, identical in deck configuration and main span length of 1500 m but differ in steel and carbon fiber reinforced polymer (CFRP) stay cables, are first designed based on the equivalent strength principle. Finite element analyses are then conducted to investigate wind-resistant performances of both schemes, including static structural behaviors, mean wind-induced deflections, buffeting responses, flutter instability, wind-induced cable resonance, vortex-induced cable vibrations and wind-induced local bending deformations of the stay cables. The results indicate that the using of CFRP cables reduces significantly the wind loads, which account for a major part in the total wind loads developed on the entire structure. As a result, mean wind-induced global deflections of CFRP scheme are notably reduced compared with the steel scheme. In terms of buffeting, results of the CFRP scheme are 12%, 14%, and 28% lower than those of the steel one in vertical, torsional, and lateral directions, respectively. No substantial difference is observed between the two schemes regarding the bridge deck flutter stability. As far as wind-induced cable resonance is concerned, the CFRP scheme is obviously superior to the steel one, exhibiting a much lower likelihood of buffeting-induced resonances due to much higher natural frequencies of stay-cables. As far as the vortex-induced vibration is concerned, however, CFRP stay-cables are less favorable than steel ones. Finally, aiming at the inherent shortcoming of CFRP cables, wind-induced bending deformations at anchorage ends are analyzed. The results show bending angles of the CFRP cables are significantly lower than those of the steel cables. With a wind speed as high as 52.97 m/s considered, CFRP stay cables experience only low-to-moderate bending angles, resulting in no significant strength reductions and posing no substantial threat to the structural safety.

Keywords: Cable-stayed bridge, Carbon fiber reinforced polymer (CFRP), Dynamic property, Wind resistant performance, Finite element analysis

1 Introduction

As a typical structure connecting two locations and enabling transportation, bridges constitute a vital component of modern transportation systems. In major engineering projects spanning valleys, rivers, or canals, the construction of long-span bridges

is indispensable. Cable-stayed bridges and suspension bridges have emerged as the primary structural systems for achieving breakthroughs in bridge spans. Compared to suspension bridges, cable-stayed bridges exhibit greater resistance stiffness, granting them superior aerodynamic advantages, while maintaining significant economic competitiveness for spans around 1400 m (Nagai et al. 2004, 2001). Globally, notable examples such as the Sutong Yangtze River Bridge (1088 m), Stonecutters Bridge (1018 m), Russky Island Bridge (1104 m), and Husutong Yangtze River Rail-Road Bridge (1092 m) have been successively constructed, repeatedly breaking records for the main spans of cable-stayed bridges (Sun et al. 2023). Even the Changtai Yangtze River Bridge, with a main span of 1176 m, is currently under construction, reflecting the trend and demand for cable-stayed bridges to enter a new kilometer-scale era.

However, the continuous pursuit of longer spans in cable-stayed bridges has also revealed numerous engineering challenges. Extensive engineering data and related studies (Hamilton 1995; Wang et al. 2025) indicate that hundreds of long-span bridges constructed worldwide since the mid-twentieth century have experienced varying degrees of corrosion in their steel cable systems. For most cable-stayed bridges with corroded cables, conventional methods such as anti-corrosion coatings and internal dehumidification have proven ineffective. Additionally, the span enhancement of long-span cable-stayed bridges is significantly constrained by material performance. Traditional reinforced concrete, due to its high self-weight, leads to nonlinear growth in girder deflection as spans increase. Moreover, the quadratic increase in cable self-weight with length exacerbates structural dead loads, creating dual challenges in design and construction (Wang et al. 2023), thereby further limiting span breakthroughs. Furthermore, as spans increase, the structural flexibility of long-span cable-stayed bridges grows, causing dense modal overlaps between cables and the structure (Almansour 2006). This heightens sensitivity to wind-induced vibrations, which can trigger parametric oscillations or internal resonance in cables. Since this phenomenon was first observed in 1984 on Japan's Meiko-Nishi Bridge (Matsumoto et al. 1992), similar occurrences have been reported in other bridges worldwide, leading to fatigue damage at cable ends and reduced service life (Chen et al. 2022).

Carbon Fiber Reinforced Polymer (CFRP) exhibits superior properties to steel cables, including lightweight strength, corrosion resistance, and low thermal conductivity (Liu et al. 2015). CFRP cables can reduce the proportion of dead loads, lower overall aerodynamic loads, and enhance cable natural frequencies. With equivalent volumes, CFRP cables weigh only 20%–25% of steel cables (Sun 2022). Research indicates that replacing steel cables with CFRP bridge cables could substantially reduce self-weight, enabling ultimate spans up to 4.4 times those achievable with steel (Liu et al. 2023). The application of CFRP not only improves structural efficiency but also reduces lifecycle maintenance costs (Liu et al. 2022; Yang et al. 2020a, b), demonstrating considerable potential for enabling span breakthroughs in super-long-span cable-stayed bridges.

More than thirty years ago, Meier proposed constructing a long-span cable-stayed bridge with an 8400 m main span to cross the Strait of Gibraltar (Meier 1987), utilizing CFRP stay cables. Since then, CFRP materials have been extensively investigated, researchers worldwide have continuously explored and refined CFRP cables with diverse structural configurations. The first CFRP cable-stayed bridge, the single-pylon Storchen

Brücke Bridge (Meier et al. 1996) with a 63-m main span, marked the initial implementation. Subsequent applications include the Herning Footbridge (Christoffersen et al. 1999), the 110-m-span Laroin CFRP Cable-Stayed Footbridge (Geffroy 2002), and China's first CFRP cable-stayed footbridge at Jiangsu University featuring a 30-m main span (Mei et al. 2015). However, the application of CFRP cables in large-span bridges has progressed slowly and remains limited. To date, CFRP cables have primarily been employed in footbridges and short-span highway bridges characterized by low loads and simple design conditions, providing minimal reference value for long-span or super-long-span cable-stayed bridges.

Recent studies have examined CFRP cable-stayed bridges from multiple angles. Research indicates that CFRP cables decrease internal forces in girders and pylons (Kremmidas 2004; Luo 2016; Zhang and Ying 2008). Under seismic loading, CFRP cable-stayed bridges demonstrate reduced displacement and internal force responses compared to steel-cable bridges, validating the seismic performance benefits of lightweight design. Aerodynamic evaluations of CFRP cable-stayed bridges reveal that material substitution does not compromise wind vibration safety. Including CFRP cables in a 1,600-m-span cable-stayed bridge reduce cross-sectional area, diminish static wind load effects, and improve lateral wind resistance, underscoring advantages in lightweight design and aerodynamic optimization (Zhao et al. 2022). Nevertheless, current research predominantly focuses on preliminary static and dynamic analyses (Ren et al. 2019; Xiong et al. 2011; Yang et al. 2020a, b). In aerodynamic studies, existing works frequently disregard cable geometric nonlinearities, approximating cables as truss models with equivalent elastic moduli (Fang et al. 2021; Zhang et al. 2018), especially for long-span bridges with extra-long cables, which introduces significant computational discrepancies (Freire et al. 2006). Additionally, CFRP cables exhibit lower ductility and shear strength relative to steel cables, making them particularly sensitive to anchorage-end bending and shear stresses during lateral deformations caused by strong winds. Quantitative assessments of these phenomena remain scarce, representing critical challenges that must be resolved prior to practical engineering implementation of CFRP cables (Meier 2012; Wang et al. 2014).

To address these challenges, this study replaces steel cables with CFRP cables in a 1500 m span steel-girder cable-stayed bridge, establishing two schemes (steel and CFRP cables) with equivalent spans. A comparative analysis of global structural performance, cable aerodynamic characteristics, and deformation behaviors is conducted to evaluate the feasibility of CFRP materials in super-long-span cable-stayed bridges from an aerodynamic perspective.

2 Structural designing of the bridge

2.1 Steel-wire stay cable scheme

As shown in Fig. 1, the general layout of the proposed cable-stayed bridge has a main span length of 1500 m, and the ratio of side span to main span is 0.448. Three auxiliary piers are design for each side span. Inverted A-shaped bridge pylons (see Fig. 2) are used, with a total height of 390 m, including standing 320 m above the bridge deck. The ratio of pylon height to main span is 0.26. Steel material is used for both stay cables and the main girder. A flat and streamlined box configuration is used for the main girder (see

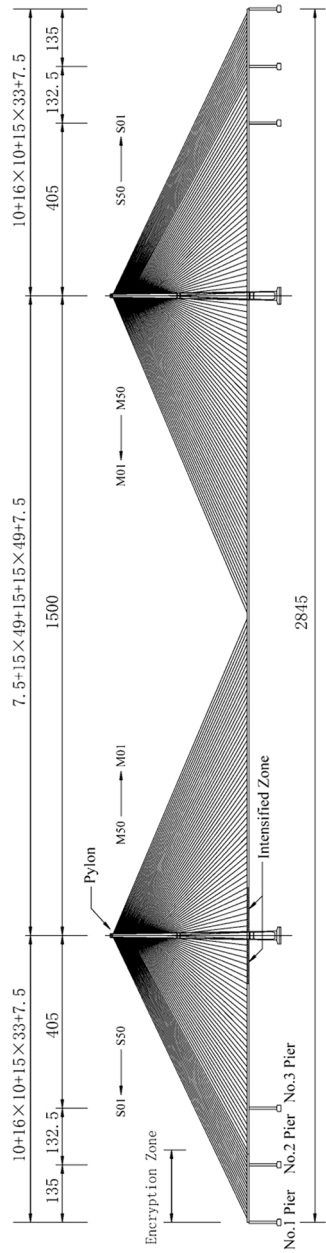


Fig. 1 General layout (unit: m)

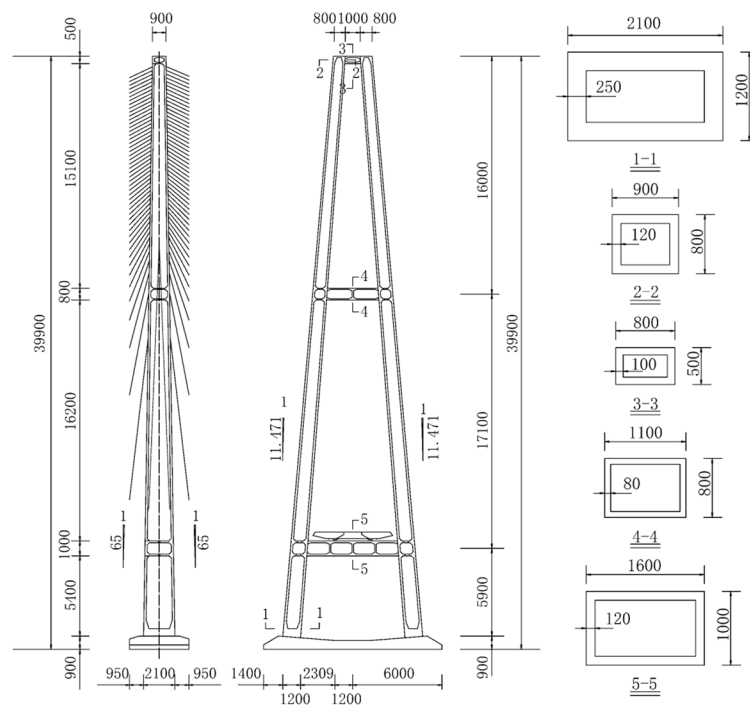


Fig. 2 Elevation of bridge pylon (unit: cm)

Fig. 3), which, equipped with wind fairings, is 53.7 m in width and 4.5 m in height. The upper and bottom plates of the girder are stiffened with U-shaped ribs. Transverse diaphragms of 10 mm in thickness, are placed within the girder with a standard spacing of 4 m. In consideration of the progressively accumulated axial force, the girder is divided into a standard zone and an intensified zone, with the latter covers 112.5 m adjacent to both sides from the both pylons. A standard horizontal spacing of the stay cables was 15 m, which reduced to 10 m in densified cable regions near the side spans. The bridge was equipped with a total of 4×100 stay cables, the cross-sectional shape of the stay cables is a circular section with dimpled surface treatment. Designated from S01 to S50 and M01 to M50, where 'S' denotes side span cables and 'M' indicates main span cables.

2.2 CFRP stay cable scheme

In general, two principles can be employed when regarding replacement of steel wire by CFRP cables. One is the so-called equivalent strength and the other is the so-called equivalent stiffness (Cai and Aref 2015). Based on these principles, the sectional area of a CFRP cable would be obtained from that of a steel wire one corresponding to the same location. When the equivalent strength theory is involved, sectional area of a CFRP cable is determined by.

$$A_{CFRP} = \frac{[\sigma]_{steel} \cdot A_{steel}}{[\sigma]_{CFRP}}, \quad (1)$$

where A_{CFRP} and A_{steel} are sectional area of the CFRP and the steel wire cable, respectively; $[\sigma]_{CFRP}$ and $[\sigma]_{steel}$ are permissible stress of the CFRP and the steel wire cable, respectively.

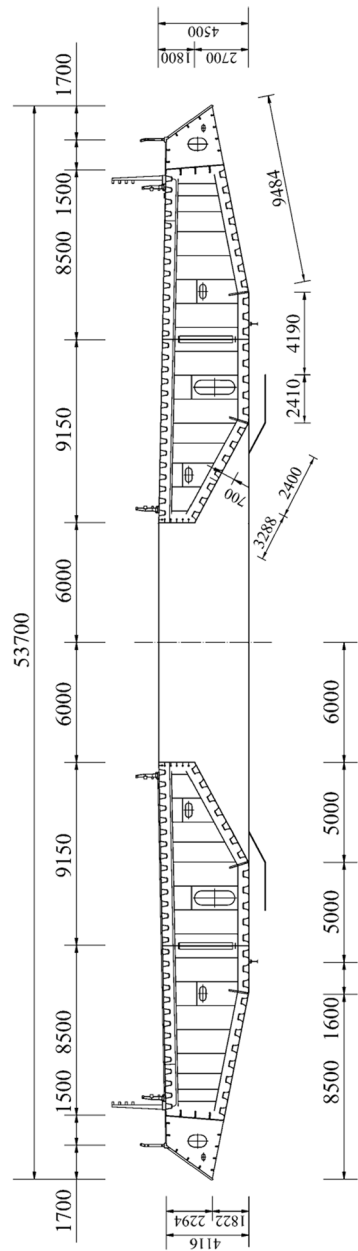


Fig. 3 Bridge deck section (unit: mm)

When the equivalent stiffness theory, the equation would be

$$A_{CFRP} = \frac{E_{steel} \cdot A_{steel}}{E_{CFRP}}, \quad (2)$$

where E_{CFRP} and E_{steel} are elastic modulus of the CFRP and the steel wire cable, respectively. For cables in a cable-stayed bridge structure, the strength, which is related directly to safety, matters much more than the stiffness does. Therefore, the equivalent strength principle is used in this work to determine the CFRP cables, with a safety factor of 2.5 designated for steel cables and a factor of 3.0 for CFRP (JTGT 3365–01–2020 2020; Feng et al. 2019).

The comparison of sectional areas and diameter of CFRP and steel cables are plotted in Fig. 4(a) and (b). It is noted that, even with the safety factor of CFRP cables being higher than that of steel cables, the sectional areas of the former are approximately 90% of those of the latter. According to the Ernst formula (Ernst 1965), the equivalent elastic moduli of stay cables using different materials were calculated as shown in Fig. 4(c). It can be observed that the reduction in the equivalent elastic modulus of steel cables is significantly influenced by cable length, horizontal inclination angle, and initial cable tension. In contrast, the equivalent elastic modulus of CFRP stay cables exhibits no notable reduction compared to steel cables and is practically unaffected by sag effects. Specifically, the M01 stay cable, located at the longest span in the mid-span, experienced the largest modulus reduction, approximately 0.58 times the original value, while CFRP cables substituted at the same location under equivalent strength and equivalent stiffness showed reductions of only 2% and 7%, respectively.

2.3 FE model and static structural behaviors

Based on the structural characteristics of the bridge, this study employs a single fish-bone model to simulate the cable-stayed bridge system. Using the relevant parameters of

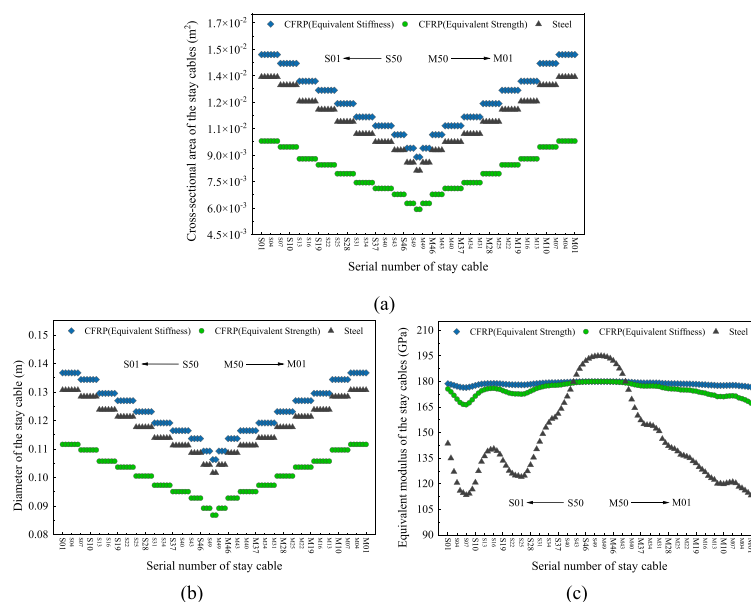


Fig. 4 Comparison of stay cable properties: **a** Cross-sectional area; **b** Diameter; **c** Equivalent modulus

the main components listed in Table 1, finite element models for two types of schemes were established using ANSYS 2021. The finite element model with hidden element shapes is shown in Fig. 5(a). In the model, the pylon and auxiliary piers were modeled using Timoshenko beam elements (BEAM188), while the main girder was simulated using Euler–Bernoulli beam elements (BEAM4). The cable-stayed structure was discretized with LINK10 elements, with each stay cable evenly divided into 10 segments to accurately capture the geometric nonlinear behavior of the cables. Details of the pylon anchorage zone in the finite element model are displayed in Fig. 5(b).

Given that this bridge is an ultra-long-span cable-stayed bridge, a fully floating system was adopted for the structure, with only transverse wind-resistant bearings installed between the pylons and longitudinal sliding bearings between the main girder and auxiliary piers. These boundary conditions in the finite element model are illustrated in Figs. 5(c) and (d), respectively. Secondary loads were equivalently simulated using MASS21 mass elements. The cross-sectional parameters of the main girder and the two types of stay cables are listed in Table 2, where the cross-sectional coordinates of the main girder align with the ANSYS global coordinate system.

Cable forces of both schemes induced by dead loads are plotted in Fig. 6. It is noticed that, with the deck contributed dead loading of both schemes being identical, forces of CFRP cables are lower in general than those of steel cables, and the reduction phenomenon becomes much more obvious for longest CFRP cables, due to their weight are far less than those of steel cables.

Dead loading induced vertical deflections at the mid-span location turn out to be 0.380 m (steel cable scheme) and 0.139 m (CFRP scheme), respectively. The deflections in both cases meet requirements designated in relevant Chinese codes, namely, less than the allowable deflection limit 3.75 m calculated from $L/400$, where L is the span length. Horizontal deflections at the pylon tops are 0.253 m (steel scheme) and 0.192 m (CFRP scheme), respectively. The largest sag is experienced by M01 cables, reaching -12.934 m (steel scheme) and -2.165 m (CFRP scheme), respectively. It is noted that the sag of CFRP cable is substantially lower than that of the steel cable, due to its low self-weight. The highest stress values in the Anchor end (pylon side) of the stay cable are respectively 424.67 MPa (steel scheme) and 459.23 MPa (CFRP scheme), less than allowable stress, namely, 0.4 times the designing strength specified by JTG D60-2015 *General specifications for design of highway bridges and culverts* (JTG D60-2015 2015). These general indices indicate that, in terms of static performances, both schemes meet requirements. However, the CFRP cable scheme turns out to be superior to the steel cable one due to smaller girder, pylon-deflections and lower static sag ratios.

Table 1 Material properties

Component	Material	Tensile strength	Modulus	Density	Poisson's ratio
Pylon	C60 Concrete	2.85 MPa	36 GPa	2549 kg/m ³	0.2
Stay cable	1860 HS steel	1860 MPa	195 GPa	8005 kg/m ³	0.3
	CFRP	2550 MPa	180 GPa	1500 kg/m ³	0.28
Main girder	Q345	345 MPa	200 GPa	8005 kg/m ³	0.3

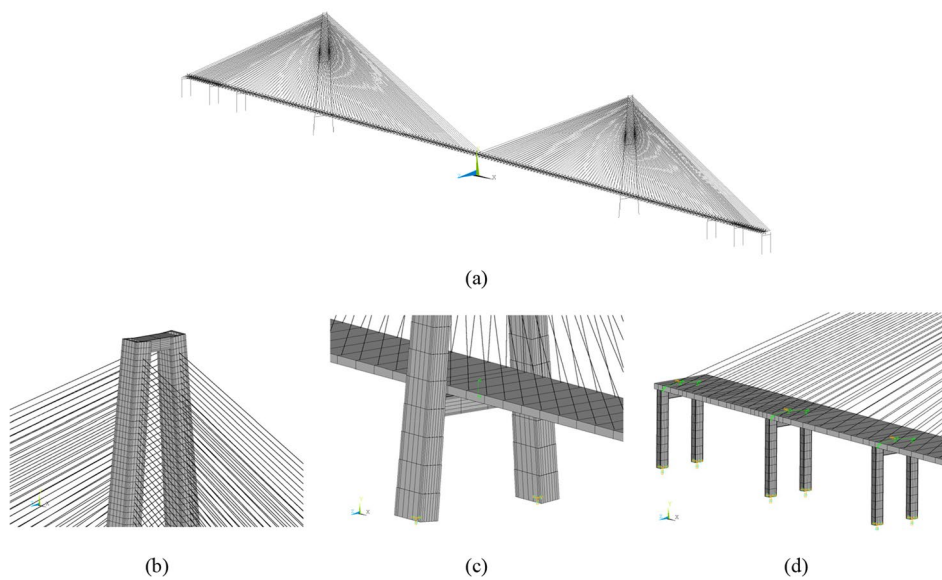


Fig. 5 ANSYS FE model: **a** isometric view; **b** auxiliary pier-girder constraint; **c** pylon-girder constraint; **d** pylon anchorage zone

Table 2 Girder and cable parameters

Section	Area (m ²)	I _{yy} (m ⁴)	I _{zz} (m ⁴)	I _{xx} (m ⁴)	Cross-sectional shape
Girder (A)	3.53	937.33	11.89	28.79	
Girder (B)	2.10	593.58	6.44	13.78	
HS steel stay cable	(0.63 ~ 1.41) × 10 ⁻²	/	/	/	
CFRP stay cable	(0.55 ~ 1.30) × 10 ⁻²	/	/	/	

3 Wind-resistant performances

3.1 Mean wind-induced deflection

According to the JTG/T 3360–01–2018 *Wind Resistant Design Specification for Highway*

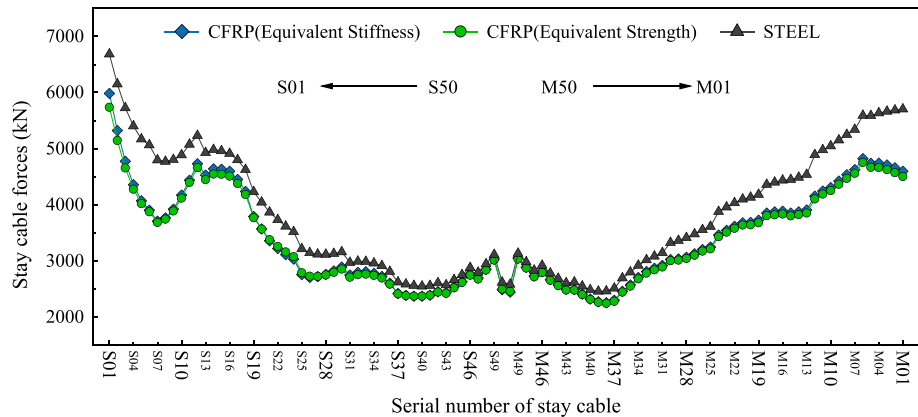


Fig. 6 Stay cable forces comparison

Bridges (JTG/T 3360–01–2018 2018), the mean wind induced aerodynamic loadings exerted on the main girder are represented by the following equations:

$$F_D = \frac{1}{2} \rho U^2 B C_D(\alpha), \quad (3)$$

$$F_L = \frac{1}{2} \rho U^2 B C_L(\alpha), \quad (4)$$

$$M = \frac{1}{2} \rho U^2 B^2 C_M(\alpha), \quad (5)$$

where F_D , F_L , M are respectively the aerodynamic drag, lift, and torque per unit length exerted on the deck axis; ρ is the air density; U represents the mean wind speed of the undisturbed upstream flow; B are the bridge deck width; C_D , C_L and C_M , functions of wind angle of attack α , are the aerodynamic drag, lift and torque coefficients, respectively. The aerodynamic coefficients (Qian 2022), obtained from wind tunnel tests by a rigid section model of the bridge deck, are plotted against the wind angle of attack in Fig. 7, and it should be noted that the static three-component force coefficients of the main girder in the wind-axis coordinate system are all referenced to the characteristic width B of the main girder.

Aerodynamic loadings on the bridge pylons, auxiliary piers and stay cables are dominated by the drag, expressed as

$$F_g = \frac{1}{2} \rho U^2 D C_d, \quad (6)$$

where F_g is the aerodynamic drag per unit length; C_d is the aerodynamic drag coefficient of the pylons, auxiliary piers, and stay cables; D is the sectional height of pylons and piers, or diameter of the stay cable. Since the aerodynamic configurations of pylons, piers and cables are regular rectangles and circulars, their related drag coefficients can

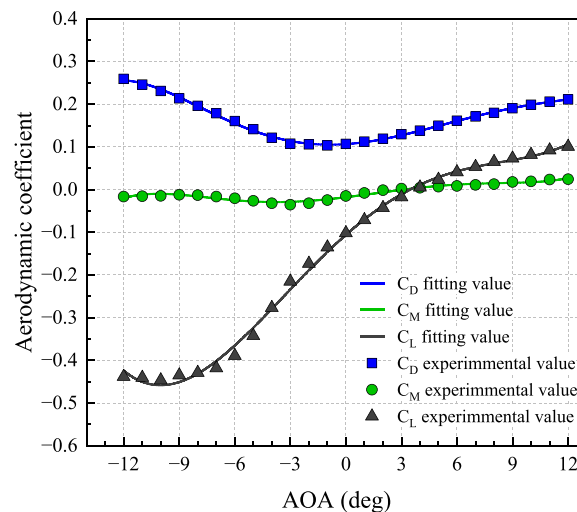


Fig. 7 Aerostatic force coefficients of deck section

Table 3 Drag coefficients of Pylon, auxiliary pier and cable

Location	Stay Cable	Auxiliary pier	Pylon
C_d	0.8	1.39	1.984 ~ 2.086

be obtained directly by turning to the specifications of JTG/T 3360–01–2018, as listed in Table 3.

According to the JTG/T 3360–01–2018 (2018), the basic wind speed U_{10} for this bridge is 38.9 m/s (derived from the 10-min average annual maximum wind speed at a height of 10 m above the bridge site, corresponding to a 100-year return period with mathematical expectation). The design wind speed U_d at the main girder height can be calculated using Eq. (7) as 52.97 m/s.

$$U_d = \left(\frac{Z}{10} \right)^{\alpha_0} U_{10}, \quad (7)$$

where Z is the reference height of the main girder; U_{10} is the basic wind speed at the bridge site; α_0 is the terrain roughness coefficient. Since the bridge site in this study falls under Category B, the value of α_0 is taken as 0.16. Three wind angles of attack (AOAs), -3° , 0° and $+3^\circ$, are covered in this study. A diagram of the FEM exerted with mean wind loadings are given in Fig. 8. Structural deflections at major locations are listed in Table 4. The longitudinal displacement at the top of the cable tower is defined as positive towards the mid-span direction, and the lateral displacement is positive in the downwind direction; for the mid-span of the main girder, vertical displacement is positive upward, lateral displacement is positive in the downwind direction, and torsional displacement is positive in the same direction as the aerodynamic torque.

It is noticed the discrepancy between the two schemes differ among wind AOAs. In general, deflections of the CFRP scheme are obviously smaller than those of the steel scheme. In terms of pylon top deflections, those of the CFRP scheme are 1% to 30% less than those of the steel scheme. In terms of the lateral main girder deflections, a quite steady discrepancy of about 25% between the two schemes is found for all wind angles of attack.

The lateral and torsional deflection distributions along the deck axis are plotted in Fig. 9. For all wind AOAs, the lateral deflections of the main girder of the CFRP scheme are larger than those of the steel scheme, and the discrepancy becomes progressively larger as the position moves toward the mid-span. The torsional deflections do not share

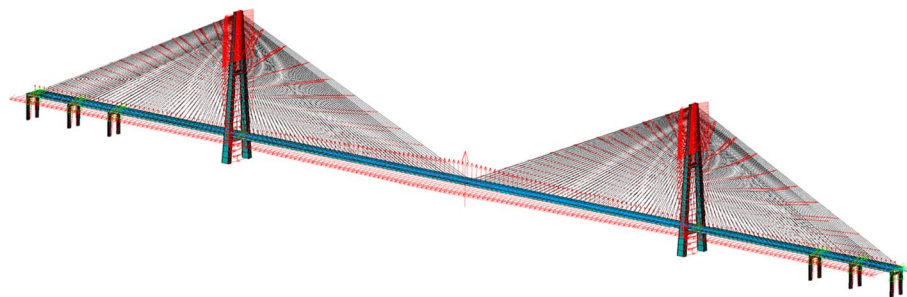
**Fig. 8** Mean wind loadings shown in FEM

Table 4 Pylon top and midspan deflections at different wind angles of attack

AOA	Scheme	Pylon top		Midspan		
		Longitudinal (m)	Lateral (m)	Lateral (m)	Vertical (m)	Torsion (deg)
−3°	STEEL	0.400	0.420	1.408	−0.247	−0.271
	CFRP	0.346	0.384	1.090	−0.489	−0.288
0°	STEEL	0.301	0.421	1.573	0.210	−0.103
	CFRP	0.246	0.385	1.254	−0.049	−0.124
3°	STEEL	0.226	0.421	1.630	0.561	0.048
	CFRP	0.149	0.419	1.310	0.292	0.017

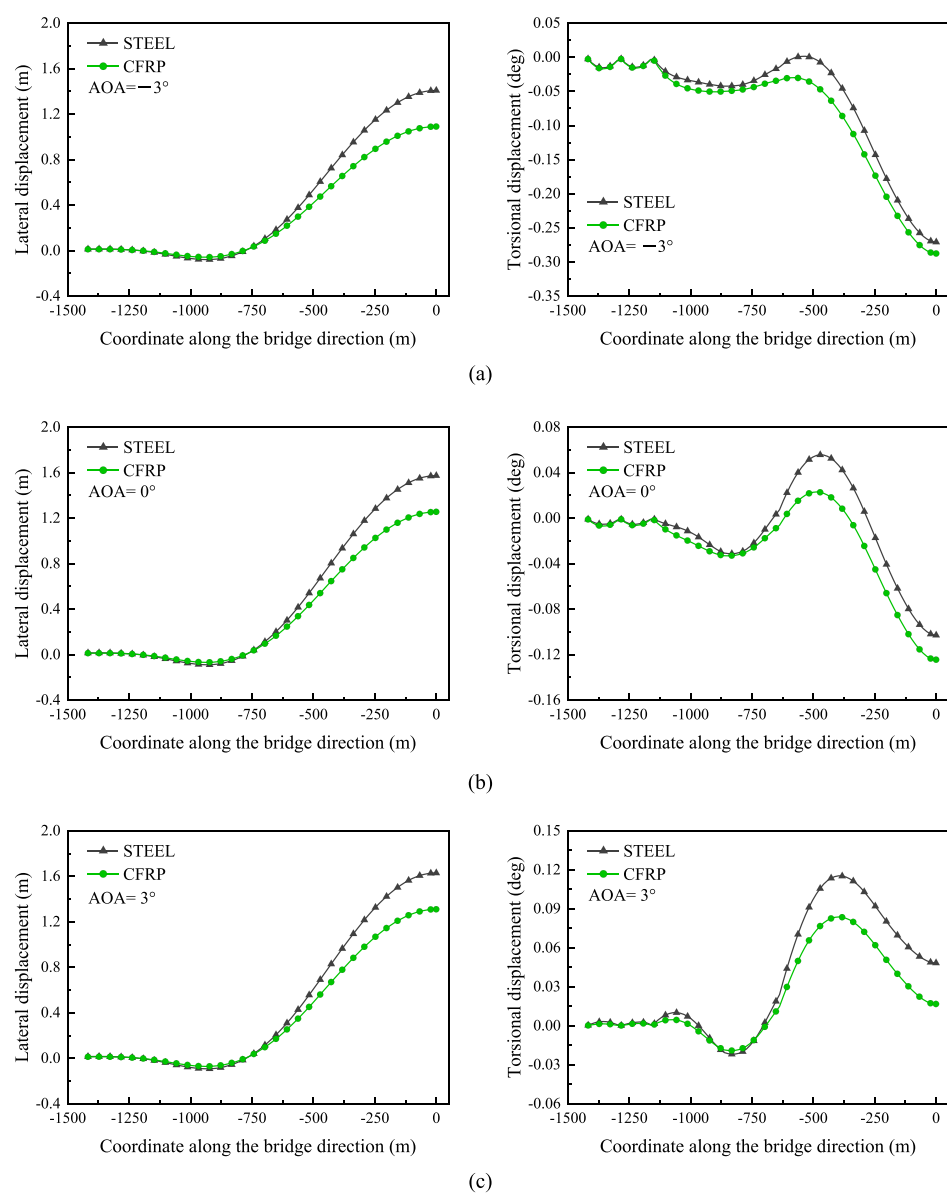


Fig. 9 Girder displacements: **a** AOA = −3°; **b** AOA = 0°; **c** AOA = 3

the same trend among different wind AOAs, with those of CFRP scheme a little larger than those of steel scheme at AOAs -3° and 0° , but smaller than the latter at AOA 3° .

In general, the discrepancy is able to be explained by the wind loadings themselves. For the super long-span cable-stayed bridge concerned in this study, the aerodynamic drag developed on the stay cables contribute a very large proportion to the total drag. Diameters of the CFRP cables, referring to Fig. 4(b), are significantly smaller than those of steel cables, which explains directly the discrepancy of lateral deflections since aerodynamic drag per unit length is linearly proportional to cable diameter. On the other hand, the CFRP cable system contributes less to the total structural torsional stiffness than does the steel cable system, resulting in slightly larger torsional deflections at wind AOAs -3° and 0° .

3.2 Buffeting responses

Refined buffeting responses are not targeted in this study, which necessitate detailed descriptions of aero-elasticities including aerodynamic damping properties, aerodynamic admittance functions, specific oncoming wind turbulence properties, etc. Instead, qualitative analyses and comparisons between the two schemes are presented. Therefore, a simplified procedure in time-domain is adopted in this study without consideration of aerodynamic damping, stiffness properties and admittance functions.

3.2.1 Turbulent wind field simulation

At the main girder height, a design wind speed of 52.97 m/s is specified. For the 190 spatial points, an improved harmonic synthesis method (Deodatis 1996; Luo and Han 2008) is employed to generate lateral and vertical turbulent wind velocity time histories with a time step of 0.25 s and a total duration of 1500 s. The design turbulence intensity for the horizontal turbulent wind speed is $I_u = 0.14$, and for the vertical turbulent wind speed is $I_w = 0.07$. The key parameters for turbulent generation and dynamic calculations in the numerical analysis are listed in Table 5.

In the dynamic time-history analysis, the Newmark- β method was employed for time integration, and Rayleigh proportional damping was assumed. The structural damping coefficients for the two design schemes were calculated based on the first vertical bending and first torsional modal frequencies of the cable-stayed bridge main girder, which are closely related to its buffeting response. The resulting target damping curves for both schemes are presented in Fig. 10. To avoid frequency aliasing issues, the upper limit frequency should be set to 4π rad/s to satisfy the constraint $\Delta t \leq 1/2f_u$, where f_u

Table 5 Parameters for turbulent wind field simulation

Item	Value	Item	Value
Simulated wind speed points	190	Upper limit frequency	4π rad/s
Simulate point spacing	15 m	Frequency segmentation	2048
Surface roughness coefficient	0.16	Time step length	0.25 s
Surface roughness height	0.05 m	Time span simulated	1500 s
Damping ratios (Steel scheme)	0.005 at 0.1556 Hz 0.005 at 0.3733 Hz	Damping ratios (CFRP scheme)	0.005 at 0.1557 Hz 0.005 at 0.4029 Hz

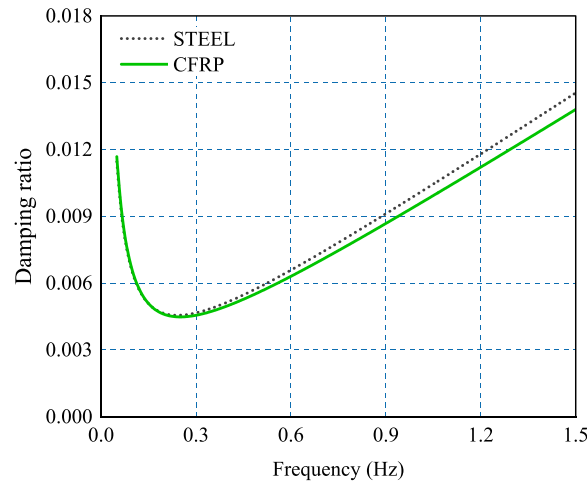


Fig. 10 Structural damping ratios

represents the upper limit frequency of turbulent wind fluctuations, Δt represents the maximum allowable time step.

The horizontal wind gusts are simulated by targeting the Kaimal wind spectrum, as

$$\frac{nS_u(Z, n)}{(u^*)^2} = \frac{200f}{(1 + 50f)^{5/3}}, \quad (8)$$

where $S_u(Z, n)$ is the power spectral density (PSD) function of the horizontal wind gust; n is frequency; Z is the height about the ground surface; $f = nZ/U(Z)$, and $U(Z)$ is the mean wind speed at height Z ; u^* denotes the shear velocity of the oncoming flow. The vertical wind fluctuations are simulated by targeting the Panofsky spectrum as:

$$\frac{nS_\omega(Z, n)}{(u^*)^2} = \frac{6f}{(1 + 4f)^2}, \quad (9)$$

where $S_\omega(Z, n)$ is the corresponding PSD function of vertical turbulent winds.

The cross-power spectral density function is

$$S_{u_1 u_2}^c(r, n) = \sqrt{S(Z_1, n)S(Z_2, n)} \cdot e^{-r}, \quad (10)$$

where

$$r = \frac{2n \left[C_z^2 (Z_1 - Z_2)^2 + C_y^2 (y_1 - y_2)^2 \right]^{1/2}}{[U(Z_1) + U(Z_2)]}, \quad (11)$$

where C_z and C_y is dependent of ground surface roughness. In this paper, C_z and C_y are taken as 10 and 16, respectively (Bucher and Wall 1992; Simiu and Yeo 2019). Typical time histories of wind turbulence are given in Fig. 11, and wind spectra obtained from simulated turbulence histories are presented in Fig. 12.

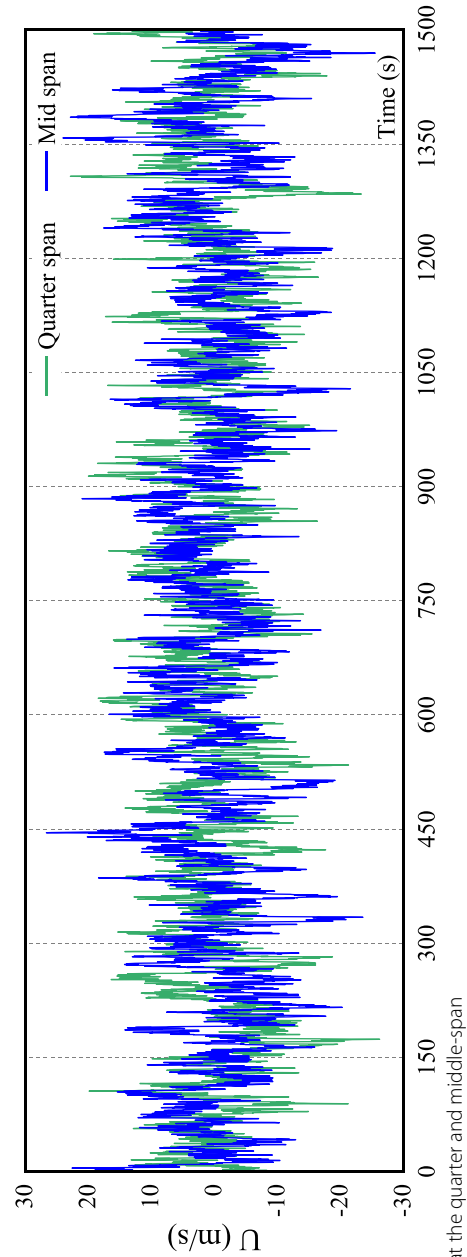


Fig. 11 Simulated $u(t)$ at the quarter and middle-span

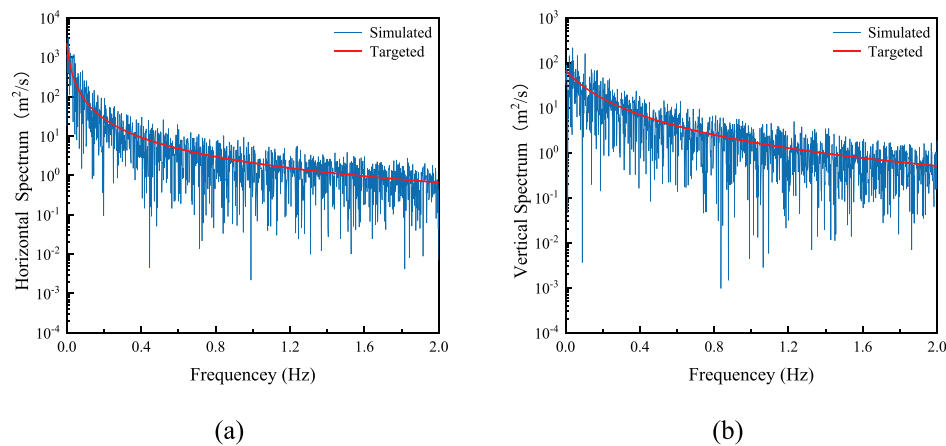


Fig. 12 Wind spectra: **a** horizontal direction; **b** vertical direction

3.3 Buffeting loads

The Davenport (Davenport 1962) expressions are adopted for describing the buffeting wind loads exerted on the bridge deck per unit span, as.

$$D_b(x, t) = \frac{1}{2} \rho U^2 B \left(2C_D \frac{u(x, t)}{U} + C'_D \frac{\omega(x, t)}{U} \right), \quad (12)$$

$$L_b(x, t) = \frac{1}{2} \rho U^2 B \left[2C_L \frac{u(x, t)}{U} + (C'_L + C_D) \frac{\omega(x, t)}{U} \right], \quad (13)$$

$$M_b(x, t) = \frac{1}{2} \rho U^2 B^2 \left[2C_M \frac{u(x, t)}{U} + C'_M \frac{\omega(x, t)}{U} \right], \quad (14)$$

where $D_b(x, t)$, $L_b(x, t)$ and $M_b(x, t)$ denote the buffeting drag, lift and torque, respectively; x denotes the bridge axis coordinate; C'_D , C'_L , C'_M are the derivatives of C_D , C_L and C_M with respect to the wind angle of attack α ; B is the reference width of the bridge deck; $u(x, t)$, $\omega(x, t)$ are the longitudinal and vertical wind fluctuations, respectively.

3.3.1 Results of buffeting responses

The lateral, vertical and torsional buffeting responses (RMS values) of the main girder are plotted in Fig. 13, where it can be seen that responses of the steel scheme in vertical and torsional directions are 12% and 14% higher than those of the CFRP scheme, respectively. The response in the lateral of the steel scheme is about 28% higher than that of the CFRP scheme, an obviously higher discrepancy than those in other directions due to the most significant reduction of buffeting loads occur in the lateral direction resulted from the decrease of diameters of CFRP cables.

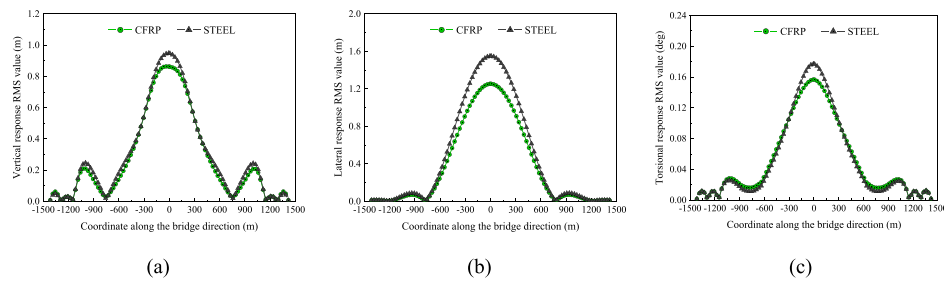


Fig. 13 Buffeting responses RMS values of main girder: **a** Vertical response; **b** Lateral response; **c** Torsional response

3.4 Bridge deck aero-elastic performances

Aero-elastic performances of long-span bridge decks are represented mainly by the flutter stability. Refined evaluation of flutter instability requires descriptions of deck aerodynamic configuration related aero-elastic properties represented by the so-call flutter derivatives (Scanlan 2000; Scanlan and Tomko 1971), the modal shape in flutter, the motion frequency, and the mass and damping properties, etc. Based on these known properties, flutter stabilities of long-span bridge decks can be obtained by well-designed sectional model tests or full bridge aeroelastic model tests in a wind tunnel. For the two schemes with an identical main girder concerned in this study, however, assessments based on the specifications of JTG/T 3360–01–2018 are sufficient to result in qualitative comparisons determining relative superiority of one scheme over another.

Modal analysis is performed and natural frequencies of major basic modes are presented, as shown in Table 6. As mentioned before, the design wind speed $U_d = 52.97$ m/s. According to the specification of JTG/T 3360–01–2018, the critical wind speed of flutter is determined by

$$U_f = \eta_s \eta_\alpha U_{co}, \quad (15)$$

where η_s and η_α are configuration coefficient and wind-angle-of-attack effect coefficient, respectively, and

$$U_{co} = 2.5 \sqrt{\mu \frac{r}{b}} \cdot f_t B, \quad (16)$$

Table 6 Natural mode and frequency

Frequency (Hz)		Modal shape description
Steel cable scheme	CFRP cable scheme	
0.0649	0.0678	Girder floating
0.0918	0.0944	Girder 1 st symmetric lateral bending
0.1556	0.1557	Girder 1 st symmetric vertical bending
0.1902	0.1905	Girder f1 st antisymmetric vertical bending
0.2233	0.2297	Pylon 1 st antisymmetric lateral bending
0.225	0.2306	Pylon 1 st symmetric lateral bending
0.2554	0.2595	Girder 1 st antisymmetric lateral bending
0.3733	0.4029	Girder 1 st symmetric torsion
0.5025	0.5240	Girder 1 st antisymmetric torsion

Table 7 Critical wind velocity of flutter

Object	Steel cable scheme	CFRP cable scheme
Shape factor η_s	1.0	1.0
Attack angle effect factor η_α	0.8	0.8
Structure to air density ratio μ	19.23	17.91
Girder section inertia radius r	16.903 m	16.903 m
Bridge girder width B	53.7 m	53.7 m
Bridge girder half width b	26.85 m	26.85 m
Girder torsional natural frequency f_t	0.373 Hz	0.403 Hz
Critical wind speed of flutter of flat plate U_{co}	174.23 m/s	197.90 m/s
Flutter critical velocity of bridge U_f	139.38 m/s	158.32 m/s

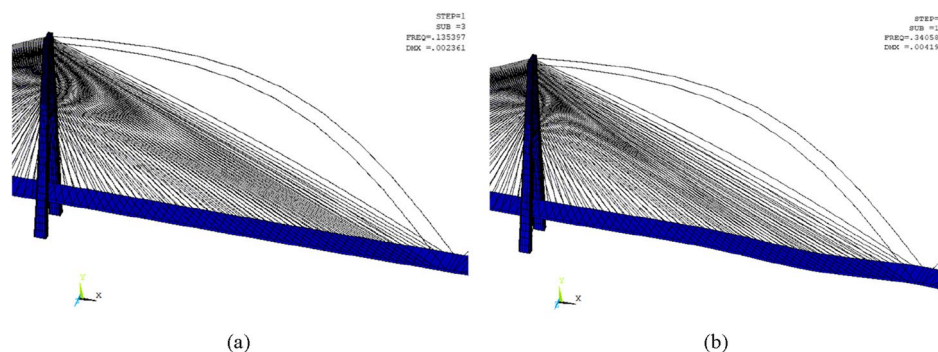
where U_{co} is the flutter threshold of a characteristic flat plate; The parameter $\mu = m/\pi\rho b^2$ represents the density ratio between the bridge structure and air, where m is the mass per unit length of the bridge (including the stay cables); r is the radius of inertia of the main girder; $b = B/2$ is the half width of the bridge deck; f_t is the natural torsional frequency of the potential flutter mode.

The results are listed in Table 7. No significant difference is found between the two schemes in terms of the deck flutter stability, although the critical wind speed of the steel scheme is slightly (approximately 13.6%) lower than that of the CFRP scheme.

4 Aerodynamic performances of stay cables

4.1 Basic dynamic properties

Since every individual stay cable is discretized into 10 bar elements, the natural vibrating properties of the stay cables are able to be obtained via modal analysis of the entire structure. Figure 14 plots the basic modal shapes of the longest stay cables of both schemes, and Fig. 15 shows the basic frequencies of all stay cables. The results of modal analysis indicate that, in the case of steel cable scheme, the basic natural frequencies vary from 0.136 Hz of the longest cable to 0.510 Hz of the shortest one. In contrast, those of the CFRP cables vary from 0.341 Hz of the longest cable to 1.430 Hz of the shortest one. The differences between the two schemes are substantial, with those of CFRP cables being 2.40 to 2.65 times those of steel cables.

**Fig. 14** Basic modal shapes of No. M01 stay cable: **a** Steel cable scheme; **b** CFRP cable scheme

This makes the CFRP-cable scheme being much more superior to the steel-cable scheme in terms of internal resonance performances, since structural internal resonance is determined by the extent of frequency matching among natural modes and, compared to steel cables, frequencies of the CFRP cables fall much more apart from the main structure's dominant frequencies.

4.2 Internal resonance

Internal resonance of stay cables is determined by the frequency-matching properties between the cables and the main structural members such as the bridge deck and pylons. Buffeting is of broadband nature. Its motion components differing in frequency are extensive, posing danger of initiating the cables' resonant vibrations (Zhang et al. 2021).

According to the wind load descriptions in Sect. 2.2, dynamic FE simulations were performed at a design wind speed of 52.97 m/s. An extensive examination of all stay cables is not intended in this study. Stay cables M03, M06, M09 and M12 in the central span are chosen as representatives, and their lateral and vertical responses are extracted and plotted for comparison (see Figs. 16 and 17). It is noticed that, with no substantial difference between the sources of exciting (lower end), the stay cables' responses exhibit significant discrepancies. The most obvious resonance occurs to the lateral responses of the steel cables, of which the cables' motions are one order of magnitude's larger than those from the lower ends (source of exciting). In comparison, the CFRP stay cables exhibit very little sign of resonance, with motions of the stay cables being the same order of those of the anchorage ends.

Figure 16(a) shows that the M03 cable, among others in the steel scheme, exhibits the most obvious resonant feature and has the largest motion amplitudes. Referring to Figs. 18 and 19, spectrum analysis reveals that, in the case of M03, corresponding to the dominant frequency, there is literally no motion component of the same frequency from the lower end (bridge deck). Figure 18(a), shows that the source of excitation actually comes from the upper end (bridge pylon). Further looking into the mechanism may demonstrate that, notwithstanding the stochastic nature of buffeting responses, it is this very tiny motion component transmitted from the upper anchorage end to the cable that result in its buffeting response characterized by an obvious resonant motion peak with a dominant frequency.

The frequency spectra of the CFRP cables also exhibit dominant and resonant peaks, as shown in Figs. 18 and 19(e) to (h). The dominant motion frequency of cable M03, 0.15 Hz, is also the dominant frequency of the bridge deck. However, the frequency is far from the basic natural frequency of the stay cable, hence it is more of a background response than a resonant one. The responses of cables M06, M09 and M12 also exhibit obvious resonant characteristics. The dominant frequencies, however, locate in a range where the energy transmitted from the main structure is too trivial to cause perceptible cable motions.

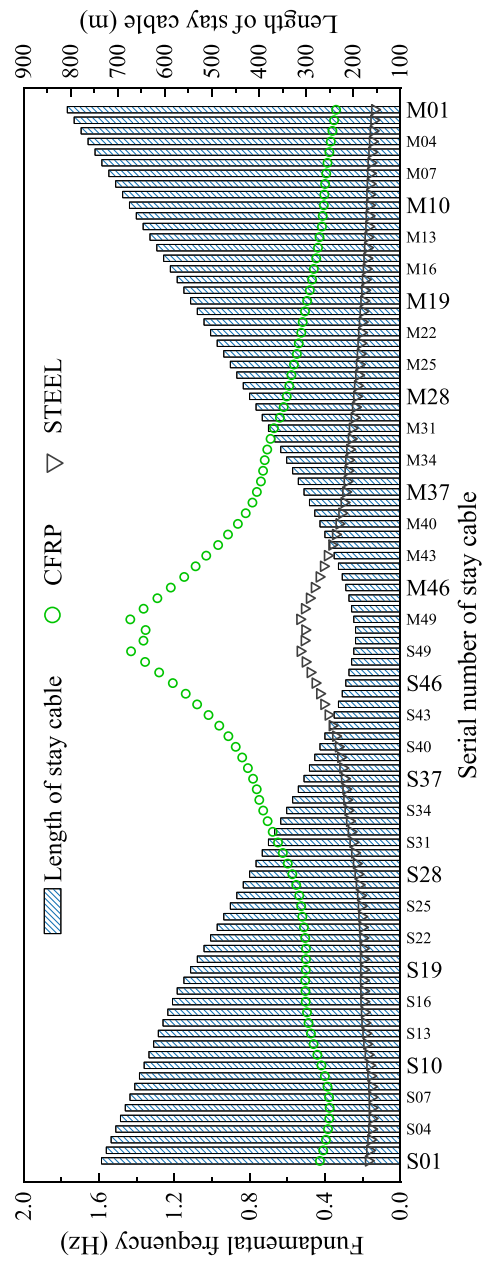


Fig. 15 Cable length and basic natural frequency

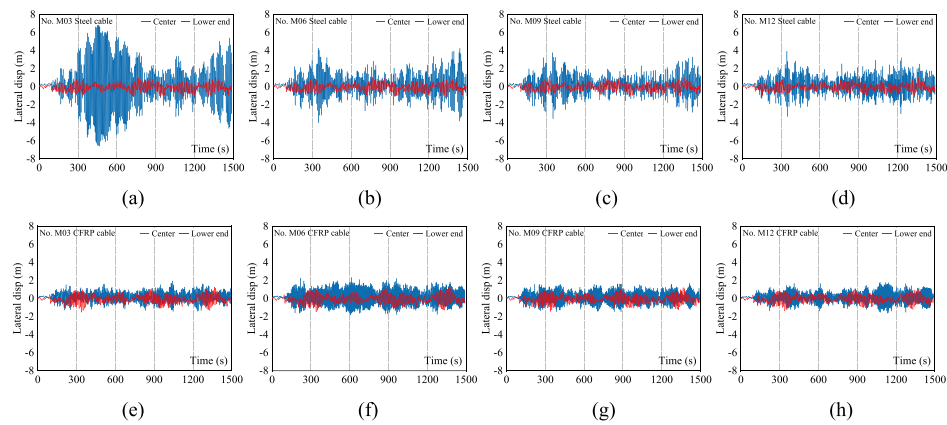


Fig. 16 Lateral buffeting responses of the stay cables for two schemes: **a** No. M03 steel cable; **b** No. M06 steel cable; **c** No. M09 steel cable; **d** No. M12 steel cable; **e** No. M03 CFRP cable; **f** No. M06 CFRP cable; **g** No. M09 CFRP cable; **h** No. M12 CFRP cable

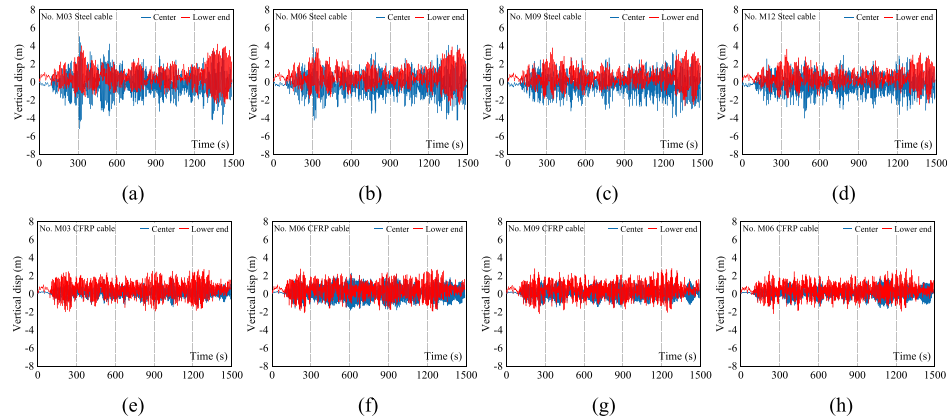


Fig. 17 Vertical buffeting responses of the stay cables for two schemes: **a** No. M03 steel cable; **b** No. M06 steel cable; **c** No. M09 steel cable; **d** No. M12 steel cable; **e** No. M03 CFRP cable; **f** No. M06 CFRP cable; **g** No. M09 CFRP cable; **h** No. M12 CFRP cable

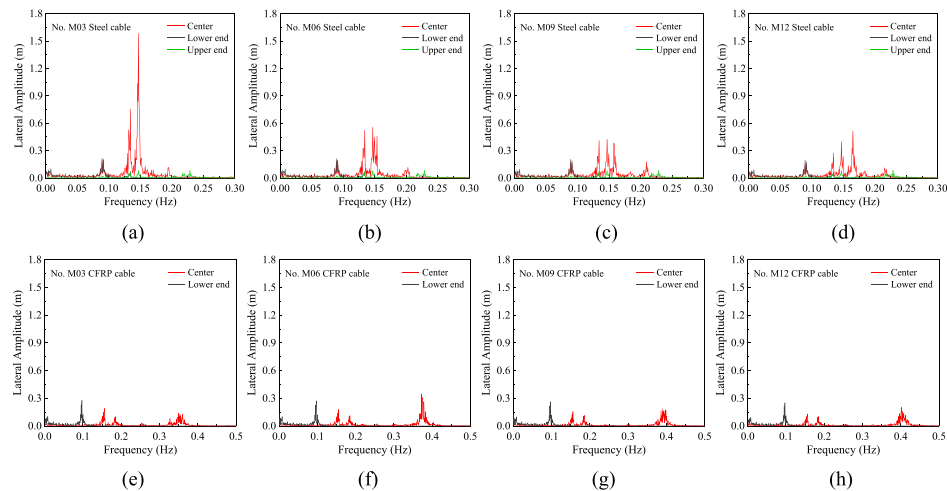


Fig. 18 Lateral response spectra of the stay cables for two schemes: **a** No. M03 steel cable; **b** No. M06 steel cable; **c** No. M09 steel cable; **d** No. M12 steel cable; **e** No. M03 CFRP cable; **f** No. M06 CFRP cable; **g** No. M09 CFRP cable; **h** No. M12 CFRP cable

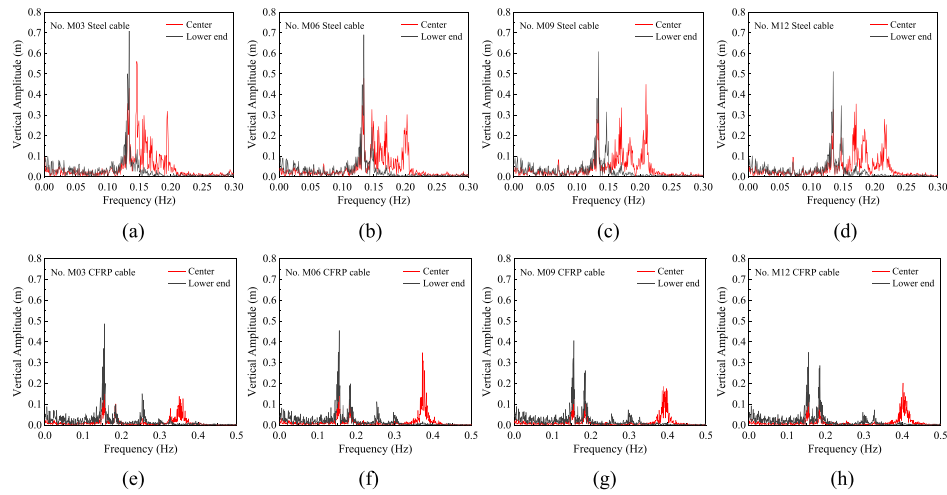


Fig. 19 Vertical response spectra of the stay cables for two schemes: **a** No. M03 steel cable; **b** No. M06 steel cable; **c** No. M09 steel cable; **d** No. M12 steel cable; **e** No. M03 CFRP cable; **f** No. M06 CFRP cable; **g** No. M09 CFRP cable; **h** No. M12 CFRP cable

4.3 Vortex-induced vibration (VIV) properties

It is well known that regular vortex shedding occurs when wind flows across a bluff body, and VIV arises when the shedding frequency matches the structural frequencies. For stay cables with circular cross sections, the frequency matching is determined by

$$S_t = \frac{f_s d}{U}, \quad (17)$$

where S_t is the so-called Strouhal number; f_s is the cable's natural vibrating frequency; d is the diameter of the cable; and U is the wind speed. In the sub-critical Reynold number range, S_t of circular cylinder is approximately 0.2. Therefore, the relation between the shedding frequency and wind speed U of each stay cable is determined by Eq. (16). In the case of steel cable scheme, where cable diameters vary from 0.089 m to 0.124 m, the lowest wind speeds initiating VIV are estimated 0.108 m/s and 0.294 m/s. In contrast, the diameters of the CFRP cables vary from 0.084 m to 0.116 m, and the corresponding lowest wind speeds initiating VIV fall within the range of 0.224 m/s to 0.646 m/s, which is significantly higher than those of the steel cable scheme. It is worth noting that these ranges for VIV are obtained from the base frequencies/modes of stay cables. VIVs corresponding to these wind speed ranges do not necessarily happen, as they are influenced by factors such as wind direction, turbulence intensity, structural damping, etc. When higher-order frequencies/modes are concerned, wind speed ranges required to initiate VIVs can be obtained in a similar fashion.

According to the specification of JTG/T 3360–01–2018, the maximum motion amplitude of VIV of stay cables is estimated to be:

$$y_{max} = \frac{0.008\sigma_{C_L}}{S_t^2 S_c} d, \quad (18)$$

where y_{max} is the maximum motion amplitude of VIV; σ_{C_L} is the standard deviation of lift coefficient, generally designated 0.45 for circular cylinders; d is cable diameter; S_c is

the Scruton number, representing combined mass-damping properties and calculated according to $S_c = m\xi_s/\rho d^2$. The higher the value of S_c is, the less likely VIV to occur. It has been reported that, with S_c being higher than 20, the possibility of VIV lock-in becomes negligible (Zhang 2018). Therefore $S_c = 20$ can be used as a threshold to determine the critical structural damping ratio that eliminates VIV phenomena. Table 8 lists the critical structural damping properties for the longest cable, the shortest cable, and the average-length cable, where the differences between the CFRP and steel cables are notable. Assuming $S_t = 0.2$ and targeting $S_c = 20$, the maximum amplitude y_{max} calculated using Eq. (18) for stay cables in both schemes is $0.0045d$.

As shown in Fig. 17, where two major trends are noticed. First, as shown in Fig. 20(a), the VIV amplitude of stay cables decreases with increasing Scruton number. For identical Scruton numbers, the CFRP cables exhibit a slightly smaller VIV amplitude (approximately 14% lower) compared to steel cables; however, this advantage requires CFRP cables to operate with a larger damping ratio than steel cables. Furthermore, as illustrated in Fig. 20(b), the VIV amplitudes of both steel and CFRP cables decrease with rising damping ratios. Notably, the steel cable demonstrates a more rapid attenuation of VIV amplitudes as the damping ratio increases. To achieve equivalent vibration control, the steel cable requires a significantly smaller damping ratio than the CFRP cable to effectively suppress VIV amplitude. Consequently, this implies that CFRP cables require a higher damping ratio to achieve the same Scruton number as steel cables. However, the CFRP cables require a significantly higher damping ratio—approximately 5.2 to 5.3 times that of the steel cables—to achieve the same maximum amplitude.

Therefore, despite CFRP cables' superior mechanical properties, their significantly lower mass compared to steel cables renders them more sensitive to external loads. This heightened sensitivity results in distinct VIV characteristics. Analysis of CFRP cables' vortex vibration performance demonstrates that when adopting CFRP materials as stay cables for long-span cable-stayed bridges, it becomes imperative to incorporate dampers or implement additional vibration mitigation measures. These precautions are essential to prevent potential VIV during operational phases.

5 The bending of stay cables

While CFRP cables offer advantages such as high strength-to-weight ratio compared to steel cables, they also exhibit drawbacks including low shear strength and weak bending resistance. Relative rotation at anchorage ends of stay cables inevitably induces bending

Table 8 Critical structural damping properties

Serial number	Scheme	Diameter (m)	Linear density (kg/m)	Frequency (Hz)	Target damping ratio
M01	STEEL	0.131 m	107.51 kg/m	0.136 Hz	0.398%
	CFRP	0.112 m	14.70 kg/m	0.341 Hz	2.122%
M25	STEEL	0.118 m	87.18 kg/m	0.217 Hz	0.399%
	CFRP	0.101 m	11.92 kg/m	0.564 Hz	2.141%
M49	STEEL	0.105 m	68.68 kg/m	0.533 Hz	0.401%
	CFRP	0.089 m	9.39 kg/m	1.434 Hz	2.109%

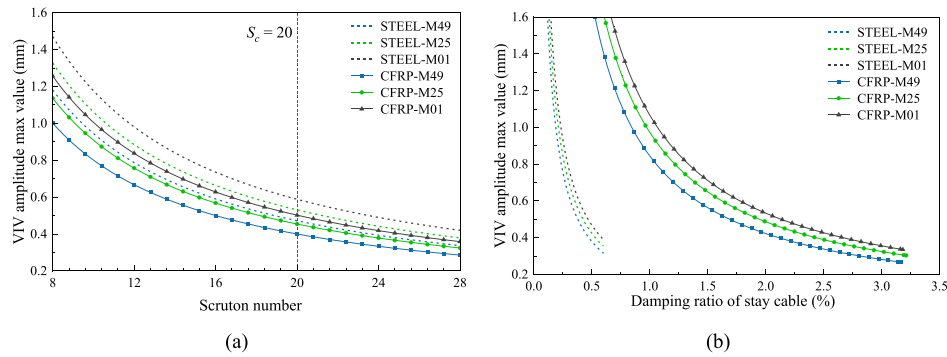


Fig. 20 Relationships between y_{max} , Scruton number, and damping ratio: **a** y_{max} versus Scruton number; **b** y_{max} versus damping ratio

deformation. In practical engineering, aerodynamic drag forces acting on cables, combined with lateral, vertical, and torsional deformations of the girder transmitted to cable anchorage zones, may collectively cause bending deformations at cable ends through coupled effects. As shown in Fig. 21(a), the wind-induced bending angles at cable anchorages can be numerically determined through finite element analysis since the model employs multi-segment truss elements to simulate stay cables.

This paper proposes a simplified method for calculating the bending angle $d\theta$ at the anchorage end of stay cables based on the discrete element model of cables. The calculation is divided into two components: one part generated by the translational displacement of cable nodes and another part induced by torsional deformation of the nodes themselves. As illustrated in Fig. 21(b), the angular deformation at the cable end is decomposed into rotation angle $d\alpha$ caused by spatial position changes of cable micro-segments through translational movement and rotation angle $d\varphi$ resulting from torsional node deformation. By performing vector superposition of these two independently calculated angular deformations, the total spatial bending angle deformation $d\theta$ at the cable end can be determined.

$$d\theta = d\alpha + d\varphi \quad (19)$$

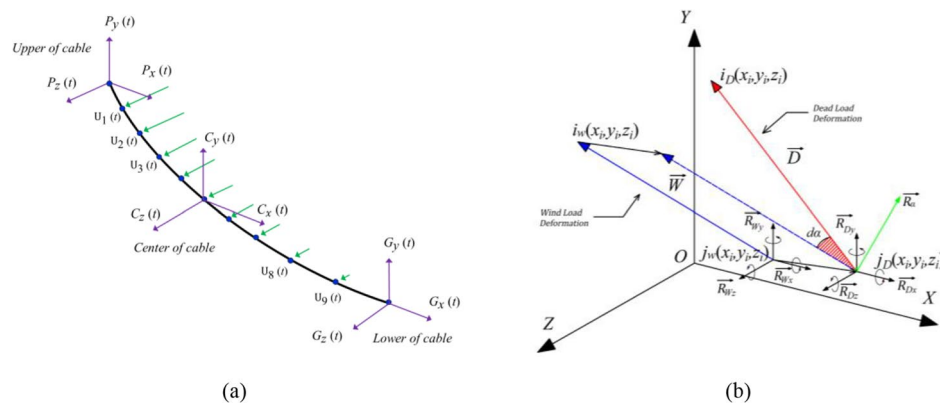


Fig. 21 The mechanism of bending deformation: **a** Diagram of overall wind-induced deformation in cables; **b** Bending deformation mechanism at cable ends

The cable micro-segment can be treated as a vector directed from the anchorage node $j(x_j, y_j, z_j)$ to the free end $i(x_i, y_i, z_i)$. Let $\vec{D}$ denote the spatial position state of the cable micro-segment in the unloaded structure, and $\vec{W}$ represent its spatial position state under external loads. The spatial angle $d\alpha$ between vectors $\vec{D}$ and $\vec{W}$, formed by translational vector displacement, contributes to the bending angle. This angular component is described using the normalized normal vector $\vec{R}_\alpha$ of the plane containing $d\alpha$:

$$\vec{R}_\alpha = \frac{(\vec{D} \times \vec{W}) \cdot d\alpha}{\|\vec{D} \times \vec{W}\|} \quad (20)$$

At the cable-girder connection node, in addition to three translational displacements, there are six degrees of freedom, including nodal torsion. The torsional displacements of the anchorage node $j(x_j, y_j, z_j)$ in three directions under the initial state and external loads are denoted as $\vec{R}_{Dx}$, $\vec{R}_{Dy}$, $\vec{R}_{Dz}$ and $\vec{R}_{Wx}$, $\vec{R}_{Wy}$, $\vec{R}_{Wz}$, respectively. Subtracting these values yields the nodal torsional angle changes in three directions at the cable end, as expressed in the following equations:

$$d\vec{R}_p = \vec{R}_{Wp} - \vec{R}_{Dp} (p = x, y, z) \quad (21)$$

By projecting the variation of the nodal torsional displacement vector $d\vec{R}_p$ onto the normal vector $\vec{R}_\alpha$ of the translational spatial angle of the cable micro-segment using Eq. (22), the contribution of torsional displacement changes to the translational rotation angle can be calculated. As shown in Eq. (23), by vectorially superimposing this contribution with the normal vector $\vec{R}_\alpha$ of the translational spatial angle and calculating its magnitude, the bending angle $d\theta$ in three-dimensional space, which accounts for torsional deformation at the cable end node, can be obtained:

$$d\vec{\varphi}_p = \frac{d\vec{R}_p \cdot \vec{R}_\alpha}{|\vec{R}_\alpha|} (p = x, y, z) \quad (22)$$

$$d\theta = \|\vec{R}_\alpha + d\vec{\varphi}_x + d\vec{\varphi}_y + d\vec{\varphi}_z\| \quad (23)$$

Bending angles (RMS value) of all stay cables due to stochastic wind loadings corresponding to designing wind speed $U_d=52.97$ m/s (at the height of deck level) are plotted in Fig. 22. It can be observed that the RMS curves of bending angles at cable ends for both schemes exhibit a trend of larger values near the mid-span. However, the CFRP cable scheme demonstrates significantly lower overall RMS bending angles compared to steel cables. Specifically, the longest mid-span cable M01 in the CFRP scheme reduces the RMS bending angle by approximately 69% at the girder end and 60% at the tower end compared to its steel counterpart.

This is primarily attributed to the smaller diameter of CFRP cables, which results in lower aerodynamic drag forces acting on the cables. Consequently, CFRP cables

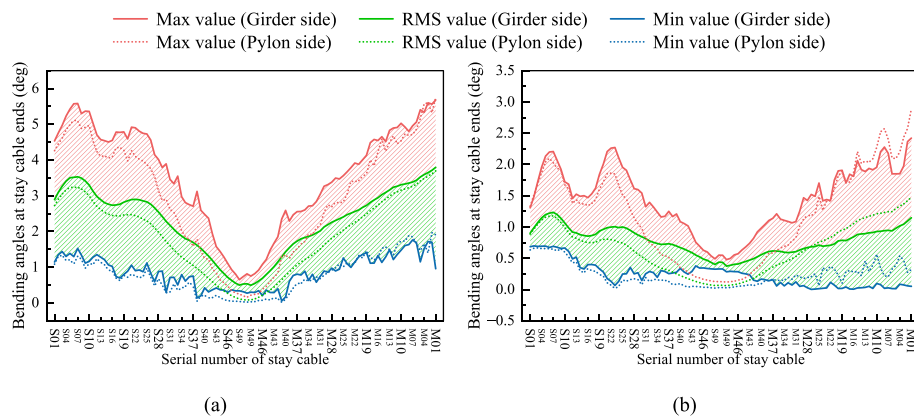


Fig. 22 RMS of bending angles of stay cables: **a** Steel stay cable scheme; **b** CFRP stay cable scheme

experience smaller lateral deformations under wind-induced loads, maintaining smaller bending angles at the cable ends. Additionally, the bending angle distribution of the CFRP cable scheme is more uniform than that of the steel cable scheme, which facilitates coordinated load-bearing between cables.

Studies have shown (Xia et al. 2022; Zhuge et al. 2014) that when the bending angle of CFRP tendons reaches 3° , their strength reduction coefficient decreases to 0.811, and this coefficient continues to decline as the angle increases. The bending tensile strength of CFRP tendons significantly decreases with larger bending angles, primarily due to their low shear strength. The shear strength of CFRP tendons is only about 5% of their tensile strength. Under significant tensile forces in a bent state, the bending region experiences pronounced shear stress. The presence of shear forces leads to a substantial reduction in the ultimate bending tensile load-bearing capacity of the tendons, ultimately causing material failure. Notably, the dominant factor leading to material failure is shear force rather than tensile force, which fundamentally explains why the bending tensile strength of CFRP tendons decreases with increasing bending angles. The expression for the strength reduction coefficient r_{sr} relative to the bending angle increment $d\theta$ is as follows:

$$r_{sr} = 1 - 0.065d\theta \quad (24)$$

Based on this, a comparative analysis was conducted between the maximum working stress of CFRP cables during the computational time history and the tensile strength reduced by bending deformation. It can be observed that the allowable stress levels of the cables exhibit a non-uniform distribution due to differences in bending angles and cross-sectional areas. As shown in the calculated results in Fig. 23, although the stress in CFRP cables fluctuates during the computational time history, the maximum instantaneous stress remains significantly below the corresponding permissible values. In this case, the minimum safety factor of 3.3 (observed in cable S01) demonstrates that the strength reduction of the cables caused by bending poses no substantial threat to the bridge structure.

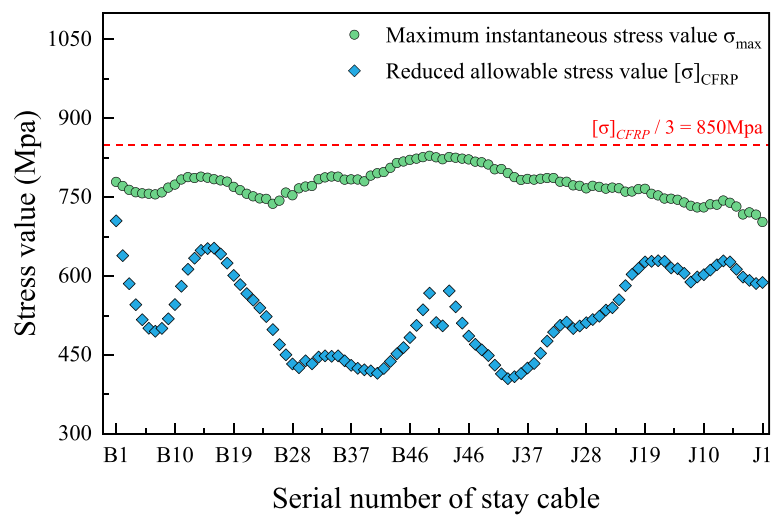


Fig. 23 Bending induced strength reduction of CFRP cables

6 Conclusions

Wind-resistant performance of two long-span cable-stayed bridges, differing in stay-cable materials, steel-wire and CFRP, respectively, is investigated and compared in this study. Through static and dynamic FE analyses, the mean wind-induced structural deformation properties, buffeting responses, VIV performance of stay-cables, and local bending of stay-cables at anchorage zones are compared. The following conclusions are drawn according to the discussions presented in this study:

- (1) Compared with the steel stay-cable scheme, mean wind-induced global deflections of the structure with CFRP stay-cables are obviously reduced due to lower aerodynamic drag developed on stay-cables. The pylon deflections of the CFRP scheme are 1% to 30% less than those of the steel scheme, and the deck's lateral deflections are approximately 25% less than those of the steel scheme for all wind AOAs.
- (2) Wind turbulences induced vertical and torsional buffeting responses of the steel stay-cable scheme are 12% and 14% higher than those of the CFRP stay-cable scheme, respectively. For the lateral buffeting responses, the discrepancy between the two schemes is even higher, with the steel scheme exhibiting a 28% higher response than the CFRP one.
- (3) There is no substantial difference between the two schemes in terms of bridge deck flutter stability. However, in terms of local aerodynamic instabilities, the CFRP scheme is obviously superior to the steel scheme, with the former exhibiting far less possibility of buffeting-induced resonances of stay-cables, due to much higher natural frequencies of CFRP stay-cables compared to steel ones.
- (4) CFRP stay-cables are inferior to steel ones in terms of vortex-induced vibrating properties. Motion amplitudes of the former would be several times larger than those of the latter. CFRP cables exhibit significantly lower Scruton numbers compared to steel cables under similar damping, making them more susceptible to larger VIV amplitudes unless additional damping measures are implemented.

- (5) Bending angles of CFRP cables due to wind loadings are significantly lower than those of steel cables. Only low to moderate bending angles (maximum value less than 3.0°) are experienced by the CFRP stay cables under a combined action of mean and buffeting wind loads corresponding to a wind speed as high as 52.97 m/s, resulting in no significant strength reductions of tendons and posing no substantial threat to structural safety.

Authors' contributions

Yuanqing Nie: Writing original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. Zhitian Zhang: Supervision, Project administration, Methodology, Conceptualization. Jiadong Zeng: Writing original draft, Visualization, Methodology, Investigation, Funding acquisition.

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Data availability

The data and materials in current study are available from the corresponding author on reasonable request.

Declarations

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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